BEGINNER GUIDE

Rendering a car in Blender for a website

What rendering is and why we need it, how to do it by hand in Blender, the exact commands I ran, the exploded “take it apart” animation, and how the website turns the pictures into scrolling 3D.

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1 Nov 2026 · Car: N°77 GR Supra GT300 by vecarz (CC BY 4.0)

01 What “rendering” means

A 3D model is only **shape + materials**: the surface of the car and what each part is made of (paint, glass, rubber). It isn't a picture yet. **Rendering** is the step where the computer works out how light from the lamps and the room bounces off every surface and into a camera — and turns that into a flat image.

	Real-time rendering (in the browser)	Offline rendering (Blender Cycles)
Time per picture	about 1/60 of a second	about 30–90 seconds
How light is computed	shortcuts and approximations	simulates real light rays bouncing (path tracing)
Reflections, glass, soft shadows	rough or faked	physically correct — looks like a photo
Can the visitor rotate freely?	yes	only through the angles we rendered
Works on phones?	struggles with big HD models	yes — it's just images

A browser must draw 60 pictures per second, so it can't spend a minute on one. That's why the first AETHER car (drawn live) looked dated. Instead we **render 120 film-quality pictures ahead of time**, one every 3° around the car, and the website shows the right one for how far you've scrolled. This is how Apple and many car brands get their photographic product pages.

> Remember: the model decides how much detail exists; the rendering decides how real the light looks. You need both.

02 The whole pipeline in one picture

```
  1. Find a model         Sketchfab → check licence + detail → download .glb
          |
  2. Build a studio        Blender: car + studio light (HDRI) + 2 light strips + dark floor + camera
          |
  3. Render                Cycles on the Mac GPU → 120 PNGs (turntable) + 4 close-ups + 48 exploded
          |
  4. Shrink for the web    PNG → WebP (1920 px for laptops, 960 px for phones), ~8 MB per set
          |
  5. Play on scroll        canvas draws frame = round(scroll progress × 119)
```





Three of the 120 turntable frames: 0°, 90° and 180°.

03 Choosing a model (licences matter)

Model we looked at	Verdict	Why
McLaren P1 “from Assetto Corsa”	❌ practice only	ripped from a video game — the uploader can't license it
Bugatti EB110 (Forza parts)	❌ practice only	also game parts, NonCommercial, real brand
Khronos “Car Concept”	✅ clean (CC BY)	made by the artist, but simpler body — looks like a concept render
Toyota GR Supra GT300 by vecarz	✅ chosen	CC Attribution, made by vecarz (kadir kandemir), 357k triangles, 4K textures

- Licence must say **CC Attribution** or **CC0** (not NonCommercial) for a public site.
- The uploader must be the maker. Avoid “Forza / Assetto Corsa / GTA / Gran Turismo / ripped”.
- Look for **150k+ triangles**, separate glass/chrome/rubber materials, 2K–4K textures.
- Download the **original .glb** — converted formats can lose detail.
- Credit exactly what Sketchfab's “Copy credits” gives you (it's in the site footer).





Same studio, three models: McLaren (great but ripped), Car Concept (clean but simple), the Supra (clean and HD).

04 Do it yourself in Blender — the turntable

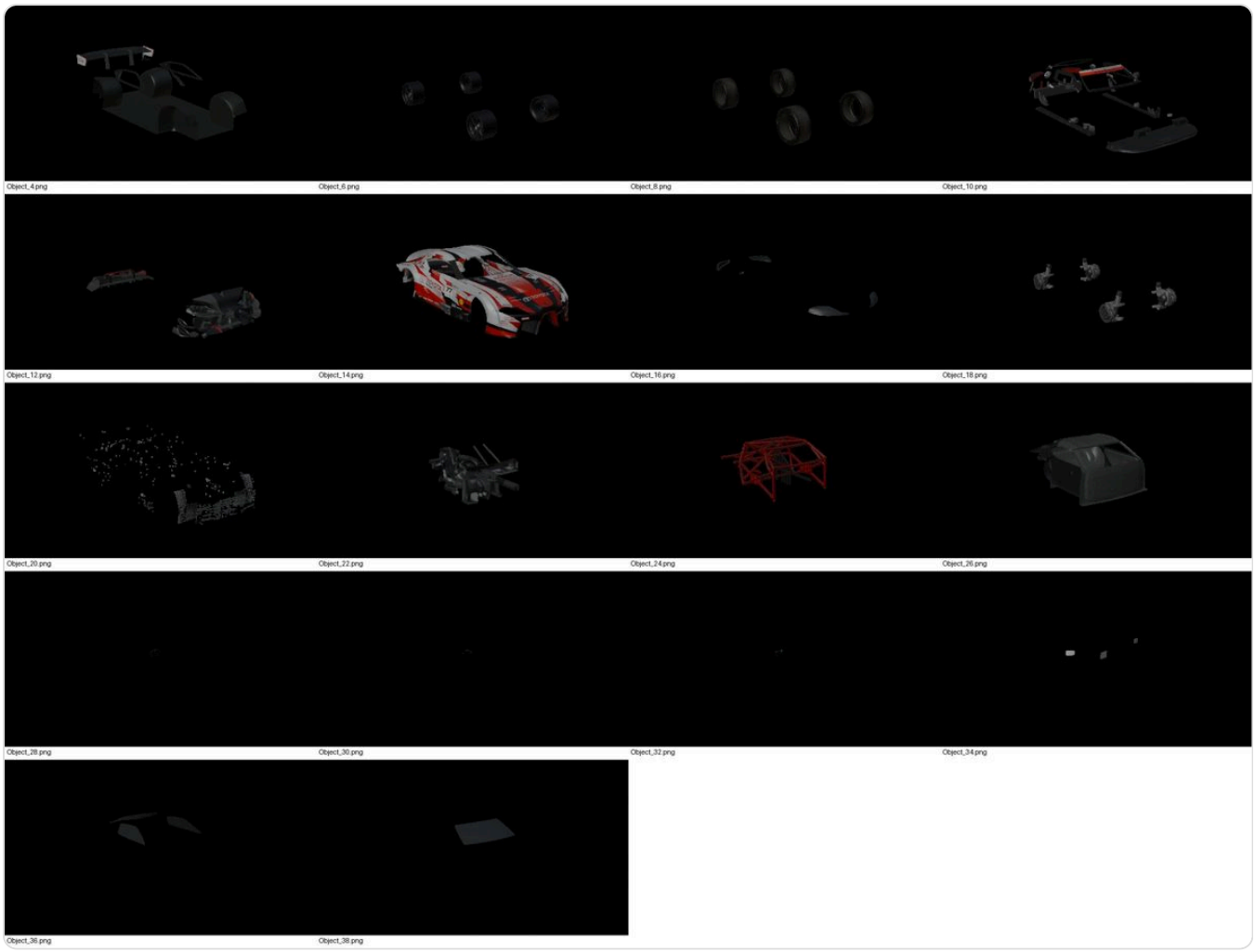
Blender is free (blender.org). These steps build exactly what my script builds. Each step teaches one idea.

Step	Where in Blender	What to do
1. Fresh scene	File → New → General	Delete the default cube, light and camera (select, press X).

Step	Where in Blender	What to do
2. Import the car	File → Import → glTF 2.0	Pick toyota_gr_supra_gt_gt300.glb. It arrives at real size.
3. Studio light	World tab (red globe) → Color → Environment Texture	Open studio_4k.hdr (Poly Haven). This lights the car and gives reflections. Strength ≈ 0.9.
4. Dark background	World node editor (Shading workspace)	Mix Shader: Light Path “Is Camera Ray” → camera sees near-black, the car still gets the HDRI light.
5. Floor	Add → Mesh → Plane, press S, type 40	Material: Principled BSDF, Base Color very dark, Roughness ≈ 0.38 (a soft reflection).
6. Light strips	Add → Light → Area	Shape Rectangle, size 0.5 × 7.5 m, Power 900 W, 4.2 m above the car. Make two, left and right. These draw the shiny lines on the paint.
7. Light only the car	Select a strip → Object tab → Shading → Light Linking	Receiver collection = a collection with just the car. The floor stays dark and moody.
8. Rim light	Add → Light → Area (violet)	Behind the car, aimed at it — a coloured edge separates the car from the background.
9. Camera	Add → Camera	Focal length 52 mm. Object Constraints → Track To → target an Empty at the car's middle.
10. Turntable	Add → Empty; parent the camera to it (Ctrl P)	Frame 1: rotation Z 0°, press I. Frame 120: Z 360°, press I. In the Graph Editor set interpolation to Linear (constant speed).
11. Render engine	Render tab	Engine: Cycles · Device: GPU (Preferences → System → Metal) · Samples 64 · Denoise ON.
12. Colour	Render tab → Color Management	View Transform: AgX · Look: Medium High Contrast (filmic, no blown-out whites).
13. Output	Output tab	1920 × 1080 · folder for frames · File Format PNG.
14. Test, then render all	F12, then Render → Render Animation	F12 renders one frame to check. Ctrl F12 renders all 120 (about 3 hours on your Mac).

> Tip: press Numpad 0 to look through the camera, and Z → Rendered for a live (slow) preview while you adjust lights.

05 The exploded “take it apart” animation



Every piece of the model rendered on its own to find out what it is: body shell, roll cage, cockpit, engine, wheels, splitter…

Part (object)	Moves	Starts at
Body shell + windows (Object_14, 36, 38, 16, 20)	up 2.7 m	0%
Roll cage (Object_24)	up 1.35 m	12%
Cockpit tub + dash (Object_26, 28–34)	up 0.45 m	22%
Engine + front bay (Object_22, 12)	forward 0.35, up 0.2	30%
Splitter / aero (Object_10)	forward 0.9 m	18%
Tyres / rims / brakes (Object_8, 6, 18)	sideways 1.25 / 0.85 / 0.45 m	8 / 16 / 24%
Floor + wing (Object_4)	stays	—

By hand: the four tyres are one object, so first split them — select it, Tab into Edit Mode, press **P** → **By Loose Parts**. Then for each group: at frame 1 press **I** → **Location**, move it (G, then Z / X and a distance), at frame 48 press **I** again. Stagger the start frames so pieces come apart one after another. Render frames 1–48 from a fixed, slightly high camera.



The fully open frame: shell on top, roll cage in red, cockpit and engine, wheels pulled out, splitter forward.

06 The commands I ran (copy and use)

I run Blender without opening its window (`-b` = background) and give it a Python script (`-P`). Everything after `--` goes to the script.

```
B=/Applications/Blender.app/Contents/MacOS/Blender
CAR=~/Downloads/assets/car/supra/toyota_gr_supra_gt_gt300.glb
HDR=~/Downloads/assets/car/studio/studio_4k.hdr
S=~/Downloads/assets/car/blender/studio.py

# 1) Turntable: frames 0–119 of 120, 64 samples, 1920×1080
STUDIO_KEEP=1 STUDIO_LENS=52 $B -b --factory-startup -P $S -- $CAR $HDR
~/Downloads/assets/car/supra/frames 0 119 120 64 1920 1080

# 2) Close-up stills (front, wheel, wing, side) at 2400×1350, 128 samples
STUDIO_KEEP=1 STUDIO_STILLS=1 $B -b --factory-startup -P $S -- $CAR $HDR
~/Downloads/assets/car/supra/stills 0 0 1 128 2400 1350

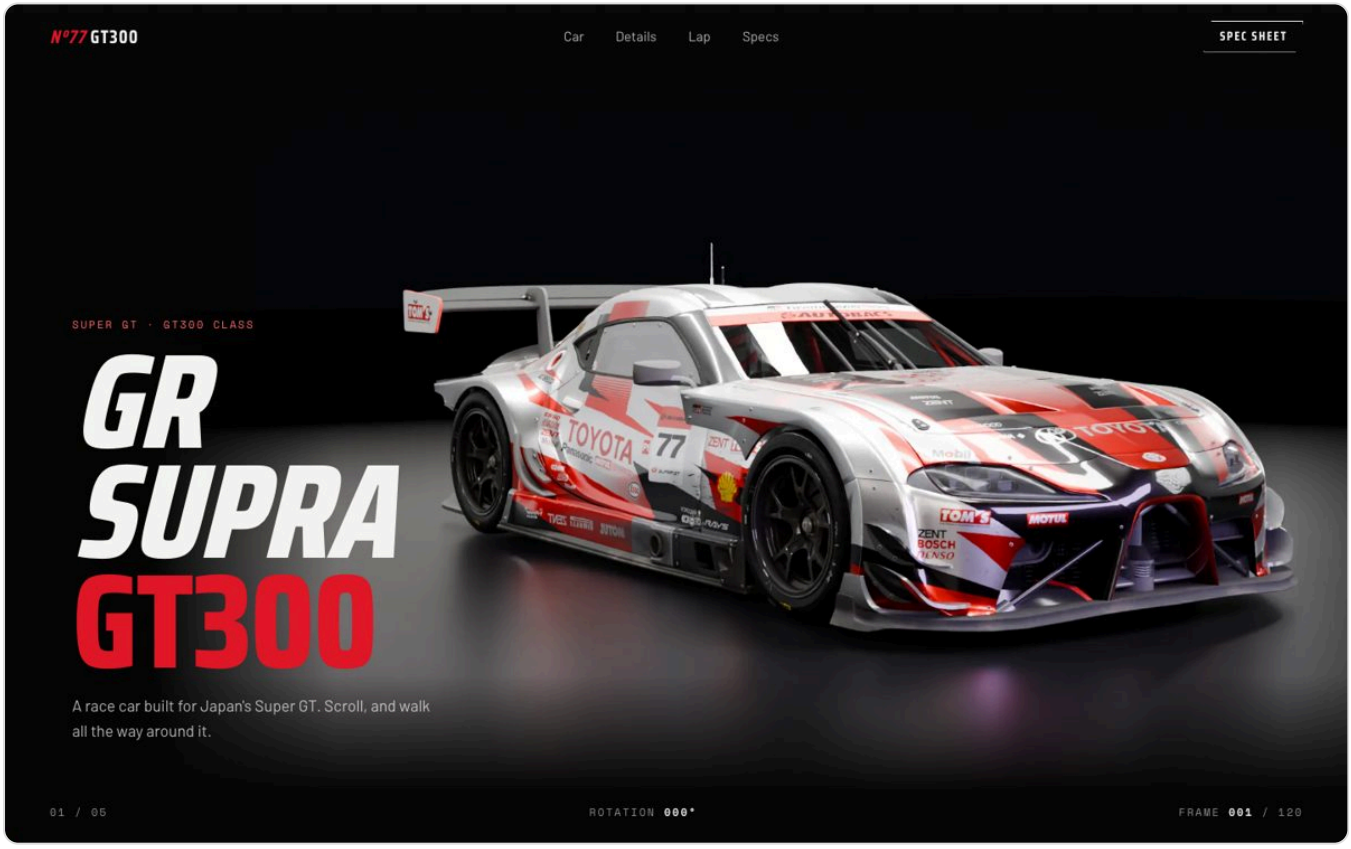
# 3) Exploded view: 48 frames
STUDIO_KEEP=1 STUDIO_EXPLODE=1 $B -b --factory-startup -P $S -- $CAR $HDR
~/Downloads/assets/car/supra/exploded 0 47 48 64 1600 900

# 4) Shrink PNGs to WebP for the site (desktop + phone sizes)
python3 ~/Downloads/assets/car/blender/to_web.py ~/Downloads/assets/car/supra/frames
~/Downloads/aether/public
python3 ~/Downloads/assets/car/blender/to_web.py ~/Downloads/assets/car/supra/exploded
~/Downloads/aether/public x_ exploded x_
```

Setting	Meaning
STUDIO_KEEP=1	keep the model's own materials (the livery); only turn windows into real glass
STUDIO_LENS=52	camera focal length in mm — bigger = closer, flatter
STUDIO_STILLS=1	render the 4 close-up shots instead of a turntable
STUDIO_EXPLODE=1	render the take-apart animation instead of a turntable
0 119 120	first frame, last frame, total frames on the circle (120 → one every 3°)
64	samples: light rays per pixel. More = less grain, slower. 64 + denoise is plenty
1920 1080	image width and height in pixels

`queue.sh` waits for the turntable to finish, then runs the stills and the exploded view automatically. You can read every line of `studio.py` — each block has a comment saying what it builds.

07 How the website plays the pictures

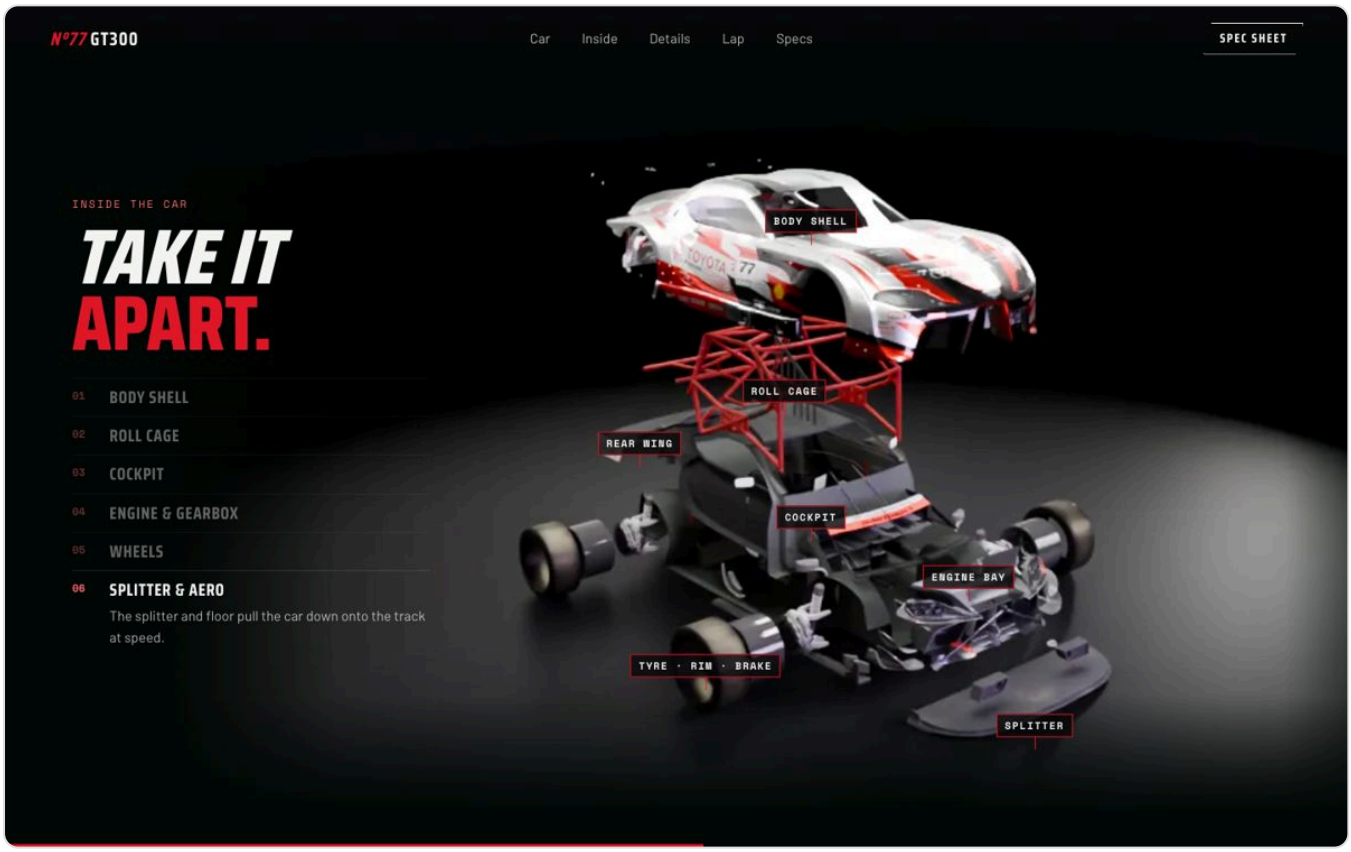


The spin: the canvas shows frame 1 of 120; scrolling turns the car.

```
// src/sequence.js (simplified)
const im = nearest(Math.round(position));      // the frame for this scroll position
g.drawImage(im, x, y, w, h);                   // draw it on the canvas
// then fade the picture's edges into the render's own background colour

// src/main.js
const p = pinProgress(spin);                   // 0 → 1 through the 560svh section
spinPos += (p * 119 - spinPos) * 0.18;         // glide toward the target frame
spinSeq.draw(spinPos);
```

- **Loading order:** every 8th frame first, then every 4th, 2nd, 1st — so the full circle works almost immediately and gets smoother as the rest arrive. The loader shows the real percentage.
- **Exploded section:** scroll 0–45% opens (frames 0→47), 45–62% holds it open while labels appear, 62–100% plays the same frames backwards so it closes again. The list on the left highlights the part being explained.
- **Phones** load half-size frames (960 px) to save data.



The exploded section: the car opens, labels point to each part, the list explains them.

08 Render times and tips

What	Time on your Mac	Size on the site
1 turntable frame (1920×1080, 64 samples)	≈ 90 s	≈ 65 KB WebP
120 turntable frames	≈ 3 hours	≈ 8 MB (desktop), ≈ 2.5 MB (phone)

What	Time on your Mac	Size on the site
4 close-up stills (2400×1350, 128 samples)	≈ 10 min	≈ 200 KB each
48 exploded frames (1600×900)	≈ 50 min	≈ 3 MB

- Test with **one frame at low resolution and 16–24 samples** before a long render.
- Grainy? Raise samples or keep **Denoise** on. Too slow? Lower resolution first (1600×900 is fine for the web).
- Keep the Mac plugged in and awake; Blender uses the GPU fully, so it gets warm — that's normal.
- If the car floats, the model has a hidden “ground shadow” sheet or stray parts — the script stands the car on its tyres.

09 Words you'll meet

Word	Meaning
Cycles / Eevee	Blender's two renderers: Cycles is realistic and slow (we use it); Eevee is fast like a game engine.
HDRI	A 360° photo of a real place, used to light the scene and give reflections.
Samples	How many light paths per pixel. More = cleaner, slower.
Denoise	An AI filter that removes leftover grain after rendering.
PBR material	Materials described physically (colour, metalness, roughness) so they look right in any light.
glTF / .glb	The web-friendly 3D file format (.glb = everything in one file).
Light linking	Telling a lamp which objects it may light (ours light only the car).
AgX	A film-like colour setting that keeps bright highlights from turning pure white.
Turntable	A camera circling an object (or the object spinning) to show every side.
Image sequence	Numbered pictures (f_000 … f_119) played like a flip-book.